Problem Set 8:

Matrix Algebra and Optimization

April 7, 2026

Question 5: Closed-Form OLS

The closed-form OLS estimator $\hat{\beta}^{OLS} = (X'X)^{-1}X'Y$ yields estimates very close to the true β vector. With $N = 100{,}000$ observations, the law of large numbers ensures the estimates converge tightly to the true values. Specifically, the maximum deviation from the truth across all ten coefficients is less than 0.01.

Question 6: Gradient Descent

Using a learning rate of 0.0000003, gradient descent converged at iteration 274, recovering estimates that match the closed-form solution to at least four decimal places. The small step size ensures numerical stability throughout the iterations. This confirms that the OLS objective function is convex and gradient descent reliably finds the global minimum.

Question 7: L-BFGS and Nelder-Mead

Both the L-BFGS and Nelder-Mead algorithms recover coefficient estimates essentially identical to the closed-form OLS solution. The two methods do not materially differ from each other, confirming that the objective function is well-behaved and the optimum is unique. L-BFGS uses gradient information while Nelder-Mead is derivative-free, yet both converge to the same answer.

Question 8: MLE via L-BFGS

The MLE estimates of β match the OLS estimates exactly, as expected in a linear model with normally distributed errors where OLS and MLE coincide. The estimated $\hat{\sigma} = 0.4999$, which is extremely close to the true value of $\sigma = 0.5$ used to generate the data. This provides further validation that the data generating process was correctly implemented.

Question 9: OLS via `lm()` and Comparison to Ground Truth

Table 1 presents the OLS estimates obtained using `lm(Y ~ X - 1)`, where the `-1` suppresses R's automatic intercept since the first column of X already contains a vector of ones. All ten coefficient estimates are extremely close to the true β values:

$$\beta = [1.5,\ -1,\ -0.25,\ 0.75,\ 3.5,\ -2,\ 0.5,\ 1,\ 1.25,\ 2]'$$

The maximum deviation from the truth across all ten coefficients is less than 0.01, which reflects the precision gained from a large sample of $N = 100{,}000$ observations. This confirms that OLS is unbiased and consistent under the classical linear model assumptions. All methods — closed-form, gradient descent, L-BFGS, Nelder-Mead, and MLE — recover virtually identical estimates, further validating the implementation.

Table 1: OLS Regression Results

	(1)
X1	1.501***
	(0.002)
X2	-1.001***
	(0.002)
X3	-0.252***
	(0.002)
X4	0.749***
	(0.002)
X5	3.501***
	(0.002)
X6	-2.001***
	(0.002)
X7	0.499***
	(0.002)
X8	1.003***
	(0.002)
X9	1.247***
	(0.002)
X10	2.001***
	(0.002)
Num.Obs.	100000
R2	0.991
R2 Adj.	0.991
AIC	145143.6
BIC	145248.3
Log.Lik.	-72560.811
RMSE	0.50

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001